

CMP5367: Artificial Intelligence and Machine Learning

Lecture 5: Classification Models

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Outline

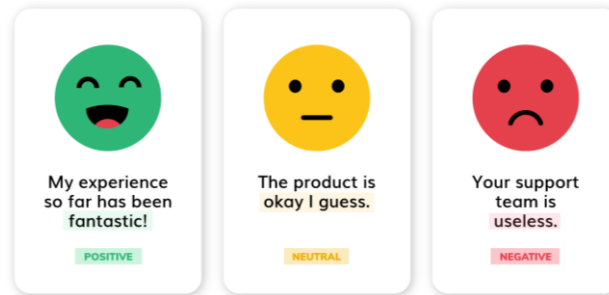
- Model 1 Decision Tree
 - ID3 Algorithm
 - Entropy
 - Information Gain
 - Model 2 Random Forest
- Model 3 Support Vector Machine (SVM)
 - Simple / Linear SVM
 - Kernel / Nonlinear SVM

In pairs decide if the following is classification or a regression problem:

(1) **Classification**



(2) **Classification**



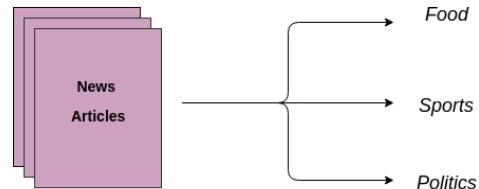
(3) **Classification**



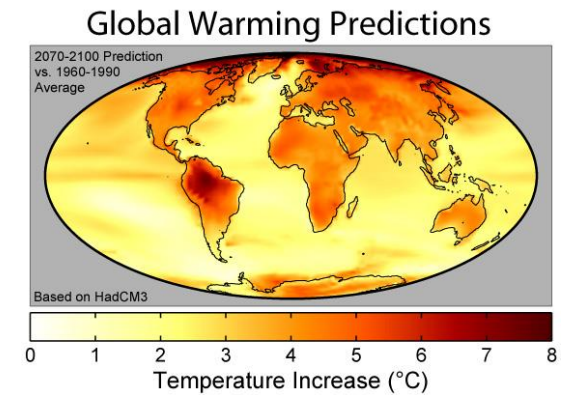
(4) **Regression**



(5) **Classification**



(6) **Regression**



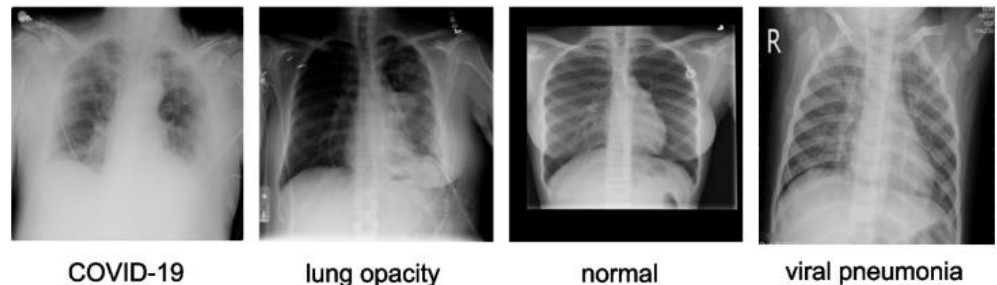
(7) **Regression**

HOUSE PRICE PREDICTION

ADVANCED LEARNING TECHNIQUES



(8) **Classification**



Remember

Classification:

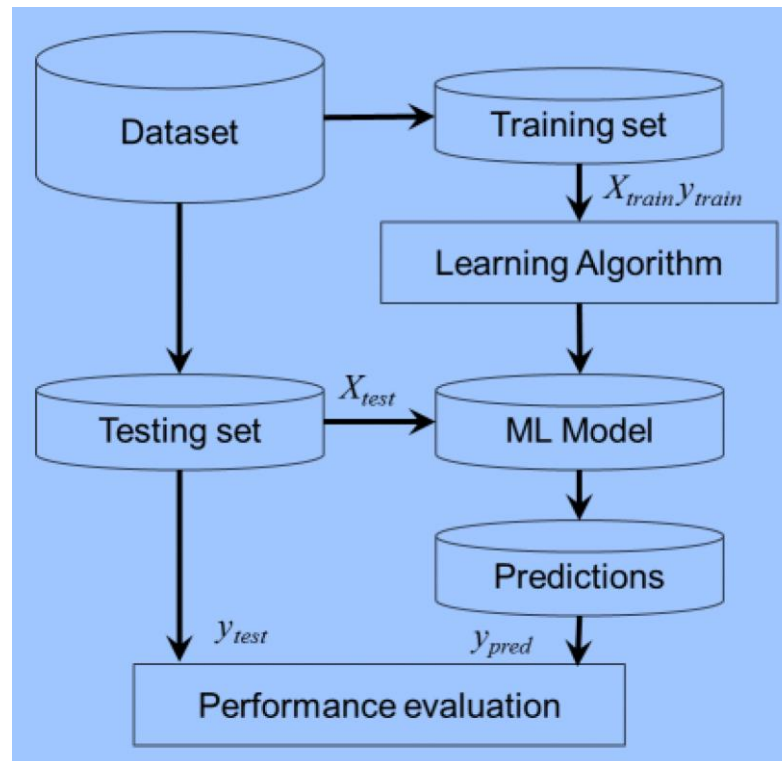
- *A machine learning task that predicts a discrete label or category for input data.*

Regression:

- *A machine learning task that estimates a continuous numerical value based on input features.*

Classification and Decision Trees

- In machine learning, Classification is a two-step process, the **learning step** and **prediction step**:
 - Model development: training data (training dataset).
 - Model evaluation (classify): new data (testing dataset).
- Decision Tree: train model that can be used to predict the class by **learning simple decision rules inferred from prior data** (training data).

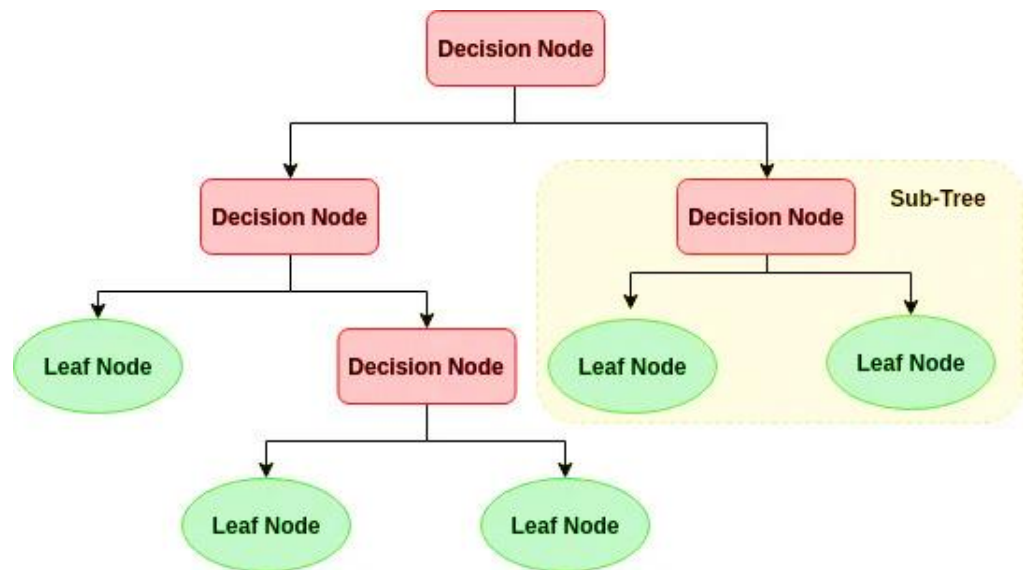


Model 1: Decision Trees

- Decision Tree algorithm belongs to the family of supervised learning algorithms. Decision tree algorithm can be used for solving regression and classification problems too.
- **Decision Nodes** involve testing a particular attribute i.e., Yes / No, True/ False, $\leq$ etc.
- **Leaf Nodes** assign classification, set of classifications, or probability distribution to instances/values.

Some Decision Tree Algorithms

- ID3 (Iterative Dichotomiser)
- C4.5
- CART
- Random Forest



Decision Trees

How Decision Trees Work:

1.Splitting Data:

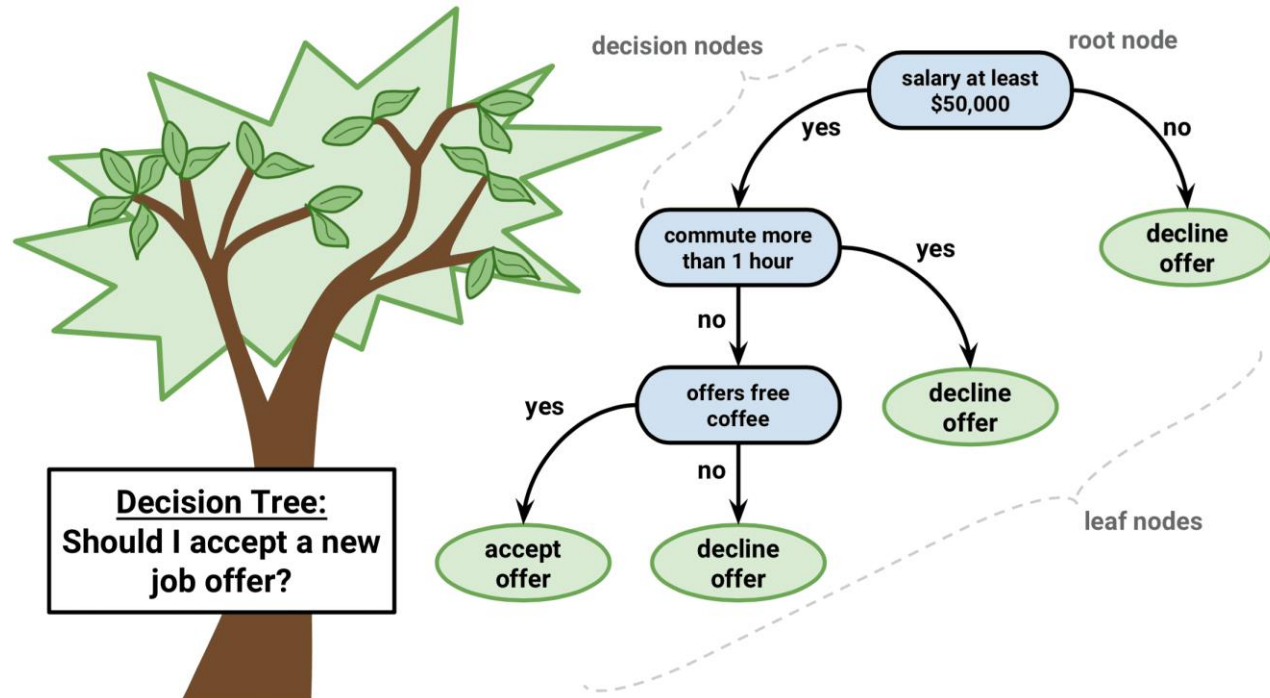
- At each decision node, the dataset is split based on a chosen feature.

2.Recursive Process:

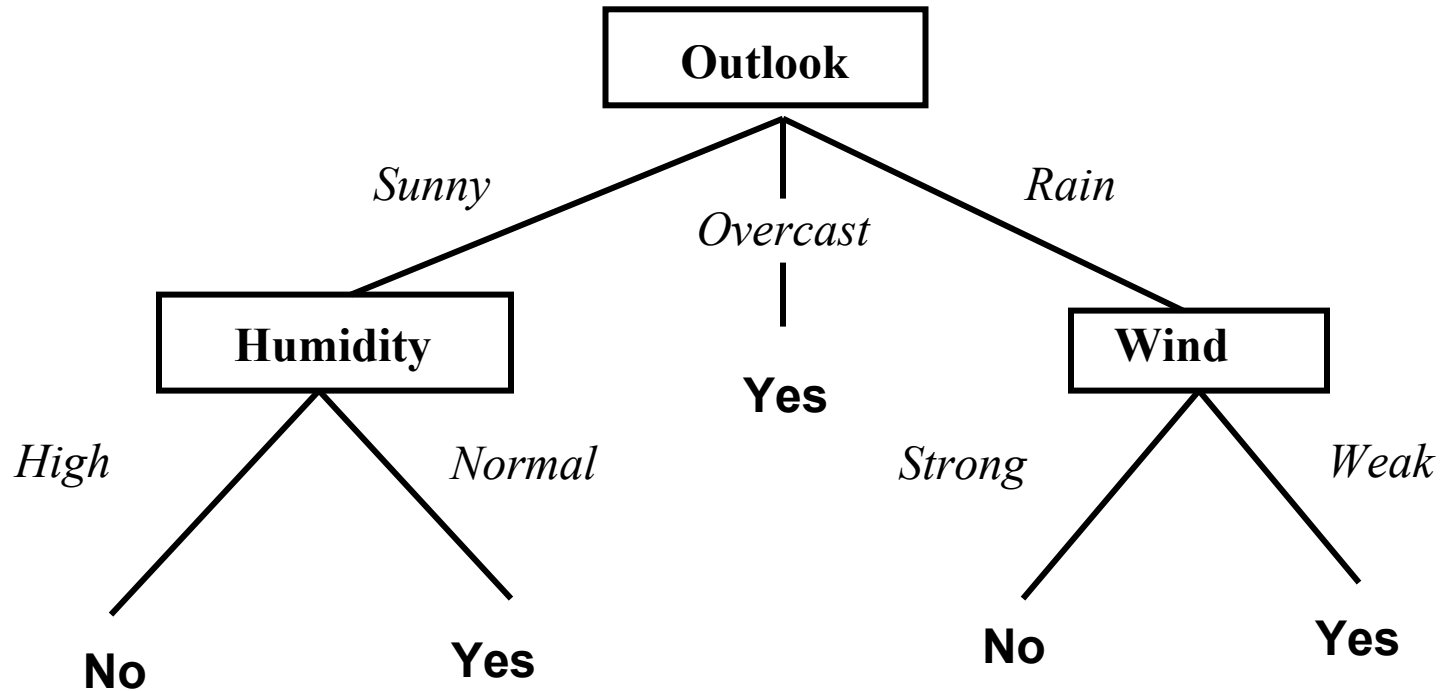
- Continues recursively until a stopping criterion is met, creating a tree structure.

3.Decision Making:

- At leaf nodes, the final decision or prediction is made.



Decision tree representation



A Decision Tree for the concept *PlayTennis*

Outlook	Temperature	Humidity	Wind	Play_Tennis
Sunny	Hot	Normal	Strong	???

Appropriate Problems for Decision Tree Learning

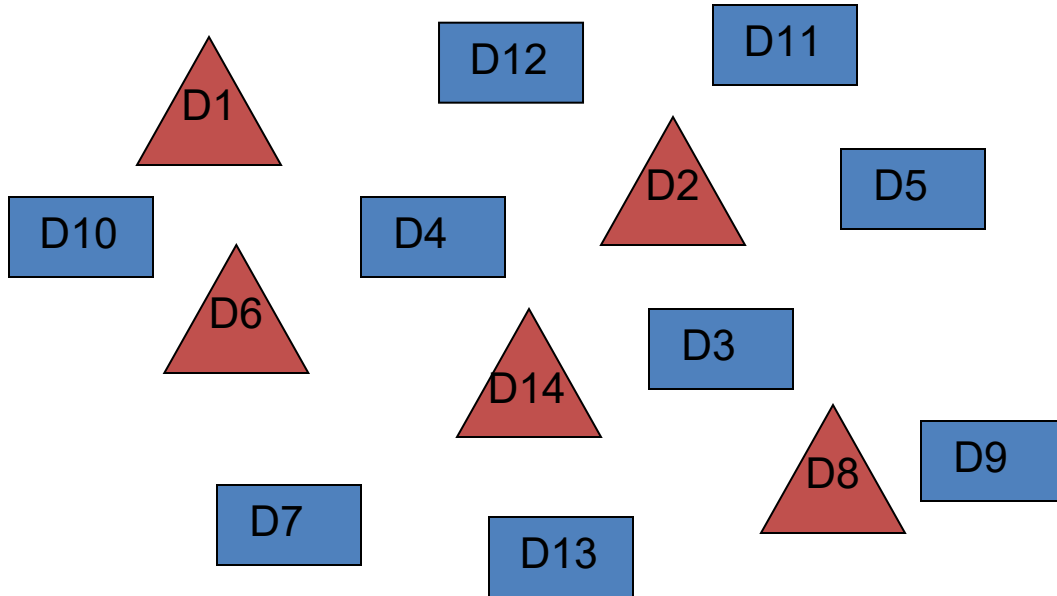
- Decision trees can handle both categorical (spam/non-spam) and numerical (1,0) features.
- The target function has **discrete output values** (e.g. classes such as: [intrusion / no intrusion], [high risk, moderate risk, no risk])
- Decision tree learning is particularly well-suited for a variety of problems, especially when interpretability, simplicity, and transparency are crucial.

ID3 Algorithm

- ID3 (Iterative Dichotomiser 3) is an influential algorithm for building decision trees in machine learning. It was developed by Ross Quinlan and is particularly designed for classification tasks.
- ID3 is a top-down, recursive approach that aims to divide the dataset into subsets based on the values of different attributes.

ID3: The Basic Decision Tree Learning Algorithm

For the dataset at the beginning (play  or not to play  ?)



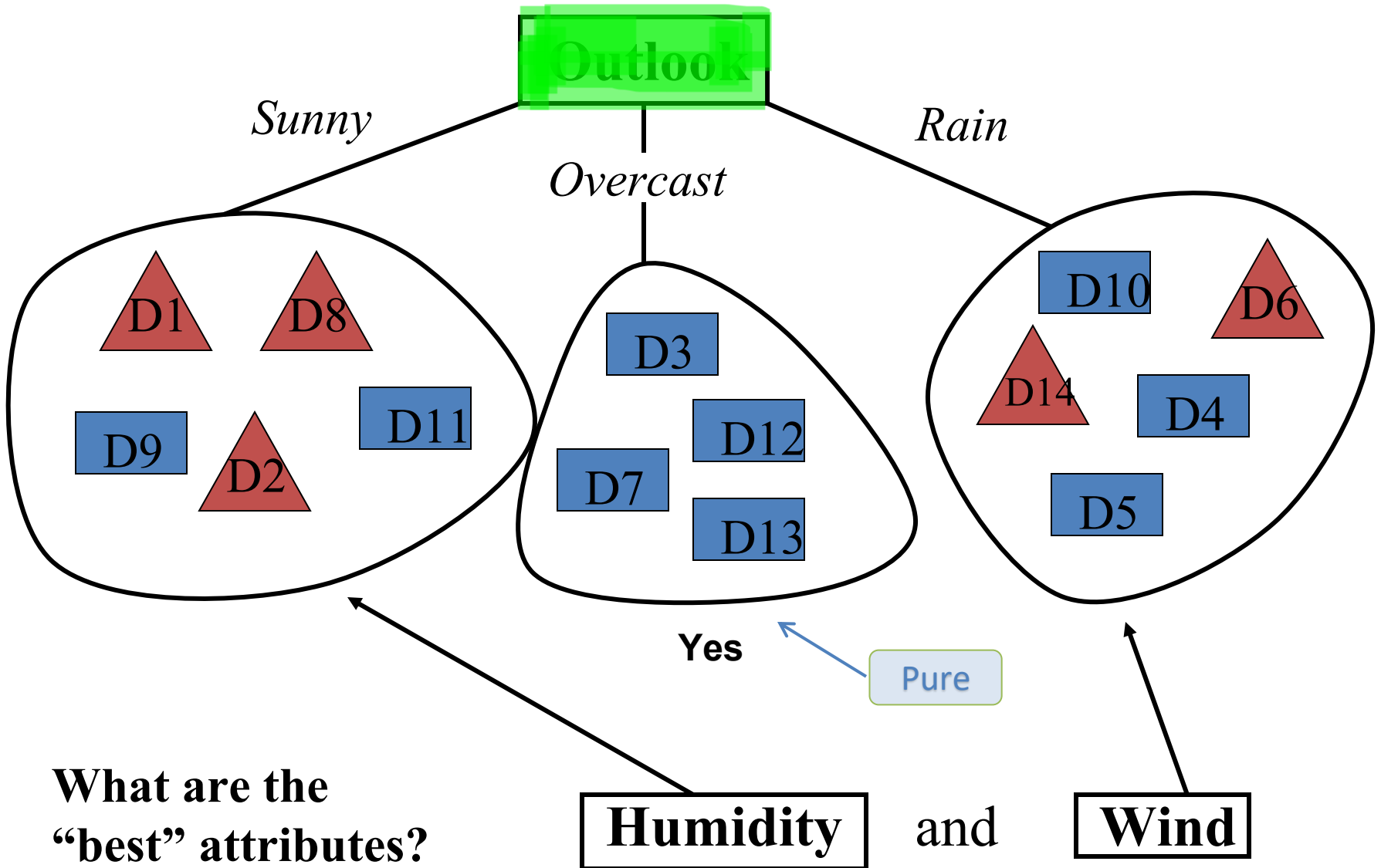
What is the “best”
Attribute to begin
the tree with?

Answer:
Try Outlook

[“best” = with highest
information gain]



ID3: The Basic Decision Tree Learning Algorithm



Decision Tree Sorting Game: Watch or not to watch Lion King?

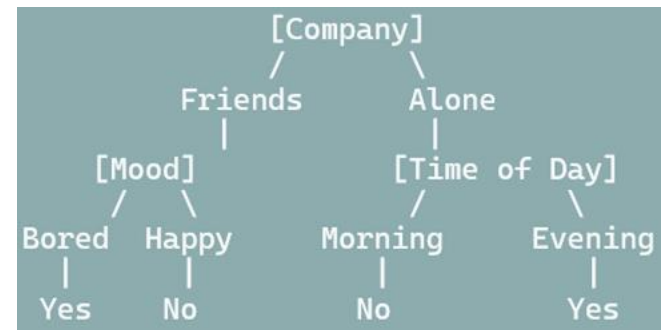
Mood	Time of Day	Company	Weather	Watch Lion King?
Nostalgic	Evening	Friends	Rainy	Yes
Happy	Morning	Alone	Sunny	No
Sad	Evening	Alone	Cloudy	Yes
Bored	Afternoon	Friends	Cloudy	Yes
Happy	Afternoon	Friends	Sunny	No



In Pairs:

1. Build a decision tree to decide whether to watch or not to watch Lion King
2. You must decide which feature to split on first, then second, etc.
3. Draw the tree on flipchart paper or whiteboards
4. Take a photo of your tree and submit [here](#)

Example Tree:



Follow up on your decision Tree

Each group selects a spokesperson to explain:

1. Why you chose each split?
2. Which features were most informative?
3. Where they encountered ambiguity?

What is the goal in constructing the decision tree?

The goal is to **decrease uncertainty** or **disorders** to be able to reach a decision.



*how do I know what should be the root node?
what should be the decision node?
when should I stop splitting?*

What is Entropy?

“**Entropy**” measures **uncertainty** or **impurity** in a dataset. It tells us how mixed the classes are in a group of examples.

Example:



4 votes No

5 votes Yes

Uncertainty: we cannot decide because we don't have a majority vote??

Why It Matters:

In decision trees, we use entropy to decide which feature to split on. A good split **reduces entropy**, meaning it makes the data **more pure** — closer to one class per group.

Entropy Formula

$$Info(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

- p_i : The probability of class i in the dataset (e.g., if 3 vote yes out of 10, then $p_{yes} = 0.3$).
- $\log_2(p_i)$: The logarithm base 2 of that probability, which measures the "surprise" or information content.
- The negative sum ensures entropy is a positive value.

Intuition Behind It

Imagine a dataset where **all examples are of class A (Yes, we will watch Lion king)**:

1. $p_{yes} = 1$ (100% of the data is class A)
2. The other class probability (P_{no} , etc.) are 0

Plugging into the formula:

1. The term $1 \times \log_2(1) = 0$ (because $\log_2(1) = 0$)
2. The terms with $0 \times \log_2(0)$ are treated as 0 (by convention, since $0 \times \log_2(0)$ is undefined but approaches 0)
3. So, the entire sum becomes:

$$\text{Entropy} = - (0 + 0) = 0$$

No uncertainty = no entropy. (I am sure all will watch the movie)

Remember:

High Entropy = High uncertainty (hard to decide)

Low Entropy = Low uncertainty (easy to decide)

Connecting back to Decision trees

- We choose to split the data on a feature that **will reduce uncertainty (low Entropy)**, in other words, will give us the **maximum Information gain (IG)** about the task
- **Information Gain** tells us how much **uncertainty is reduced** when we split the data using a particular feature.

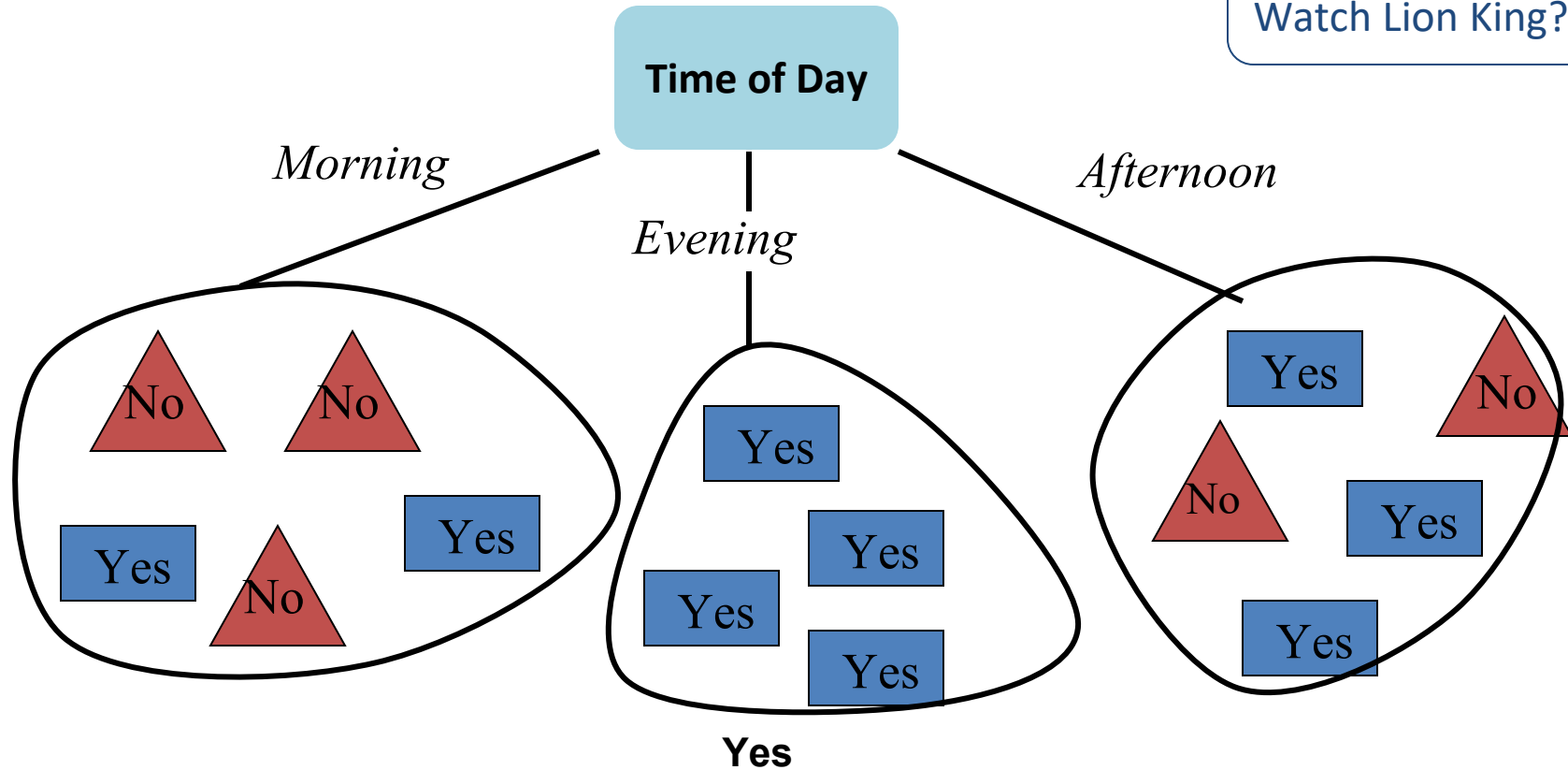
Formula:

Information Gain = Entropy(before split) – Weighted Entropy(after split)

- The **higher the IG**, the more effective the feature is at separating the classes.
- Information Gain (IG) is directly derived from **entropy**—it measures how much **entropy decreases** after a dataset is split using a particular feature.

Intuition behind Information Gain

Watch or not to
Watch Lion King?



Imagine you're trying to guess whether someone will watch Lion King. If you split the data by "Time of Day" and it makes the groups more predictable (e.g., all "Evening" times lead to "Yes"), then you've gained information—**reduced entropy**. So:

- **Entropy** is the baseline uncertainty.
- **Information Gain** is the improvement you get by splitting.

How does the ID3 Algorithm work?

Example: *WatchMovie*

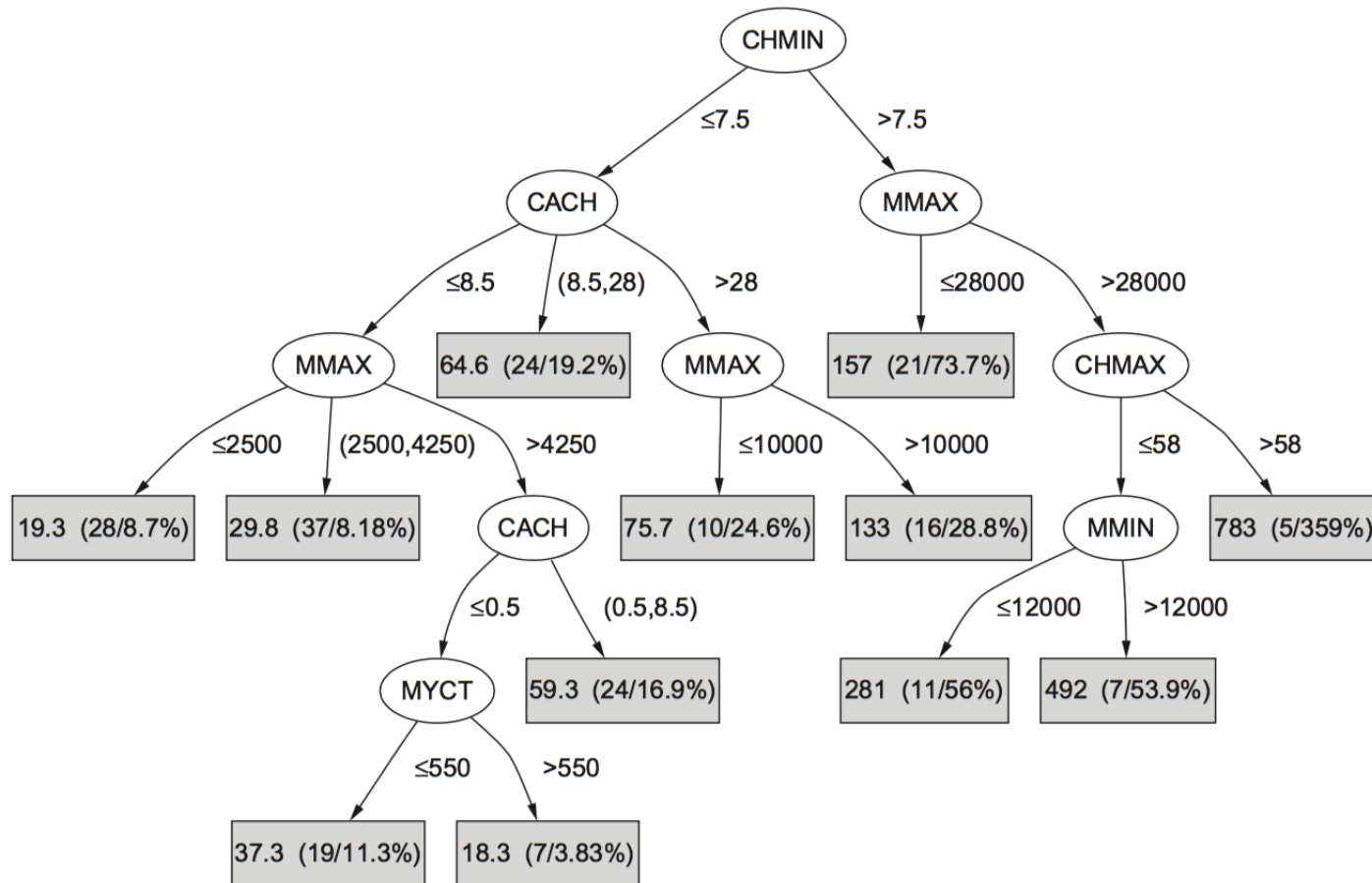
Evaluation procedures:

- Calculate the Information Gain of **attribute** – e.g. **Time of Day**
 - Step 1: Calculate **Entropy** of *WatchMovie* **before** the split (**whole dataset**)
 - Step 2: Calculate **Entropy** of *WatchMovie* **after** split based on Time of Day
 - Step 3: Calculate **Information Gain**
- Repeat the Information Gain calculation for all the other attributes (**Mood, Company, and Weather**)
- Choose the attribute with the highest information gain (IG).

When to Stop Splitting?

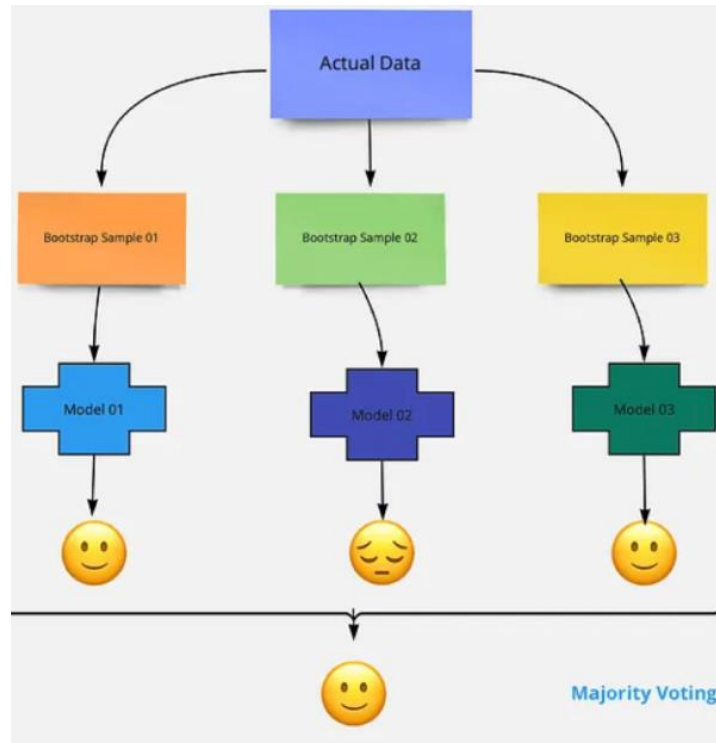
- Real-world datasets have a large number of features  a huge tree.
- Several splits can lead to overfitting  the tree will give very good accuracy on the training dataset but will give bad accuracy in test data.
- **Tips:**
 - *Set a limit on how deep the tree can grow (e.g., depth = 10).*
 - *Stop splitting if a node has too few samples (e.g., less than 5).*
 - *Stop splitting if the information gain from a potential split is below a threshold (e.g., < 0.01).*
 - *Pruning: cutting the nodes or sub-nodes which are not significant.*

Regression Tree: Example CPU data



Model 2: Random forest?

- A popular machine learning [algorithm](#) developed by Leo Breiman and Adele Cutler, *merges the outputs of numerous decision trees to produce a single outcome, suitable for both classification and regression tasks.*
- It can handle the data set containing *continuous variables*, as in the case of regression, and *categorical variables*.
- ***Random Forest is based on the idea of Ensemble Modelling:***
- It means combining multiple models to make predictions rather than an individual model.

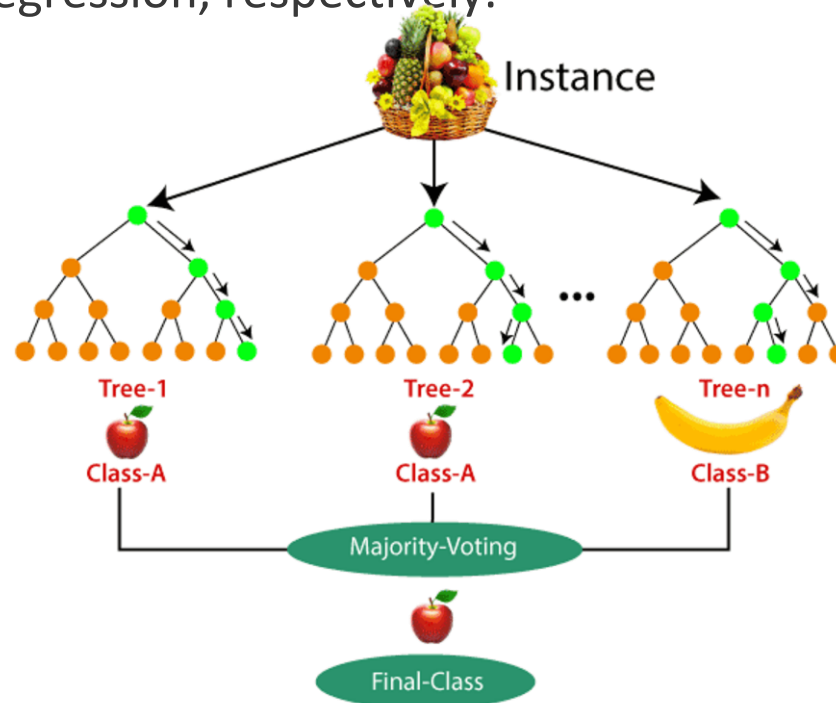


Majority Voting: The final output is determined by combining the results of all models through majority voting. The most commonly predicted outcome among the models is selected.

Random Forest Algorithm:

Bootstrap Aggregation (Bagging)

- ✓ **Step 1:** A subset of data points and a subset of features is selected for constructing each decision tree. n random records and m features are taken from the data set having k number of records.
- ✓ **Step 2:** Individual decision trees are constructed for each sample.
- ✓ **Step 3:** Each decision tree will generate an output.
- ✓ **Step 4:** Final output is considered based on **Majority Voting or Averaging** for Classification and regression, respectively.



Decision trees	Random Forest
1. Decision trees normally suffer from the problem of overfitting if it's allowed to grow without any control.	1. Random forests are created from subsets of data, and the final output is based on average or majority ranking; hence the problem of overfitting is taken care of.
2. A single decision tree is faster in computation.	2. It is comparatively slower.
3. When a data set with features is taken as input by a decision tree, it will formulate some rules to make predictions.	3. Random forest randomly selects observations, builds a decision tree, and takes the average result. It doesn't use any set of formulas.

Any Questions so far?

Quiz



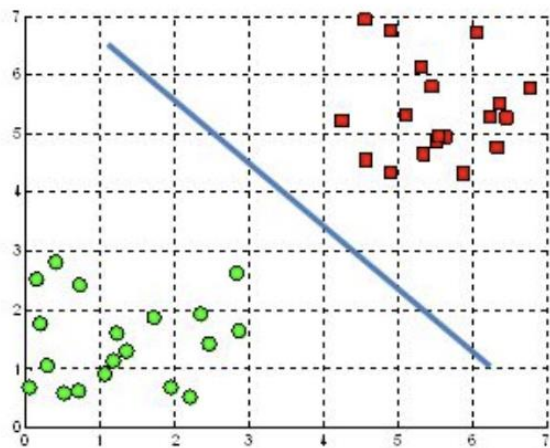
Model 3

Support Vector Machines (SVM)

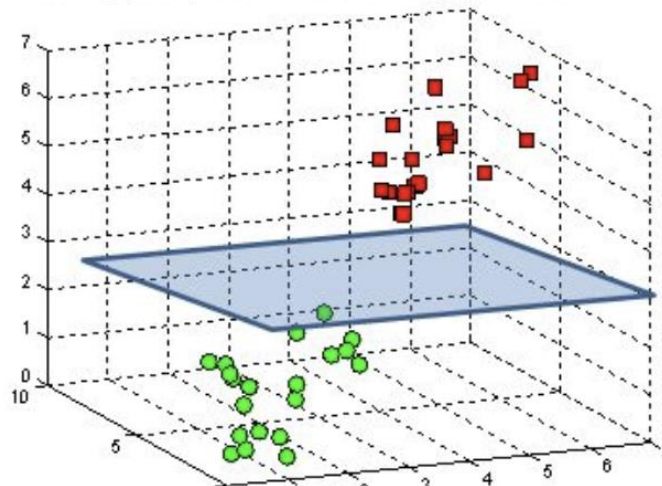
Support Vector Machines (SVM)

- SVM is a supervised machine learning algorithm used for classification and regression tasks.
- *Works best for classification with small complex data.*
- The same idea like linear regression, SVM tries to find the best separation between classes but in multidimensional space **using a hyperplane (divide a space into two half-spaces)**.

A hyperplane in $\mathbb{R}^2$ is a line

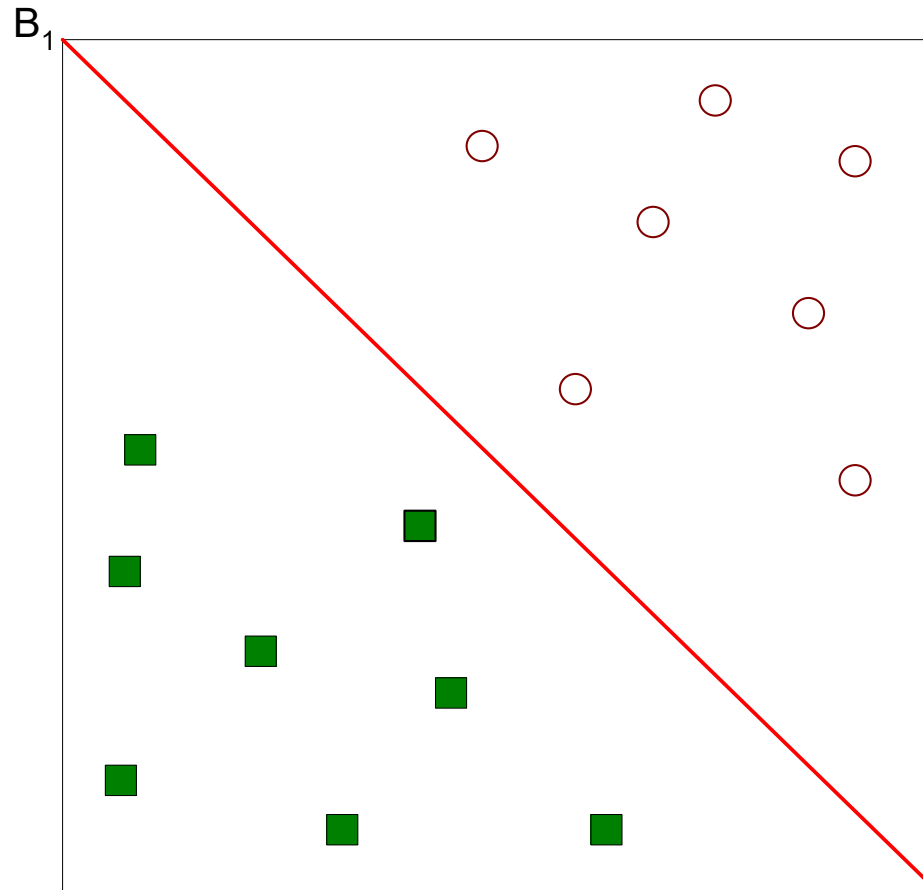


A hyperplane in $\mathbb{R}^3$ is a plane



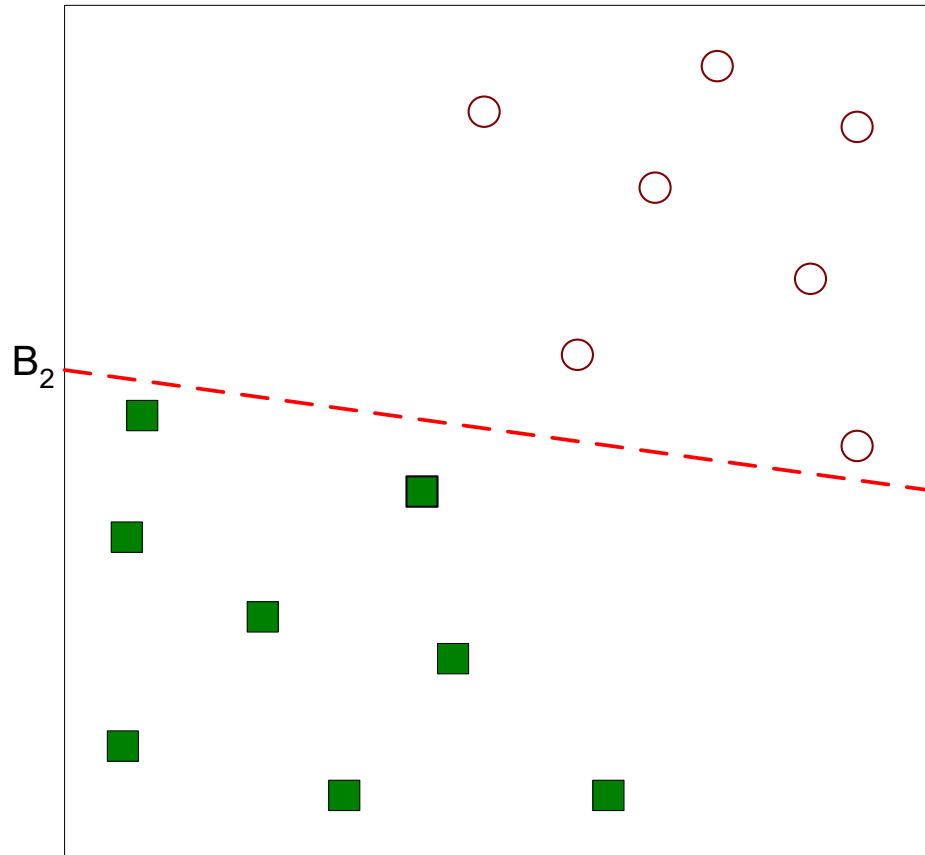
- You can either use LR or SVM but SVM works best with small complex data
- Tip: start first with LR and if it gives low accuracy try SVM with default settings.

Support Vector Machines (SVM)



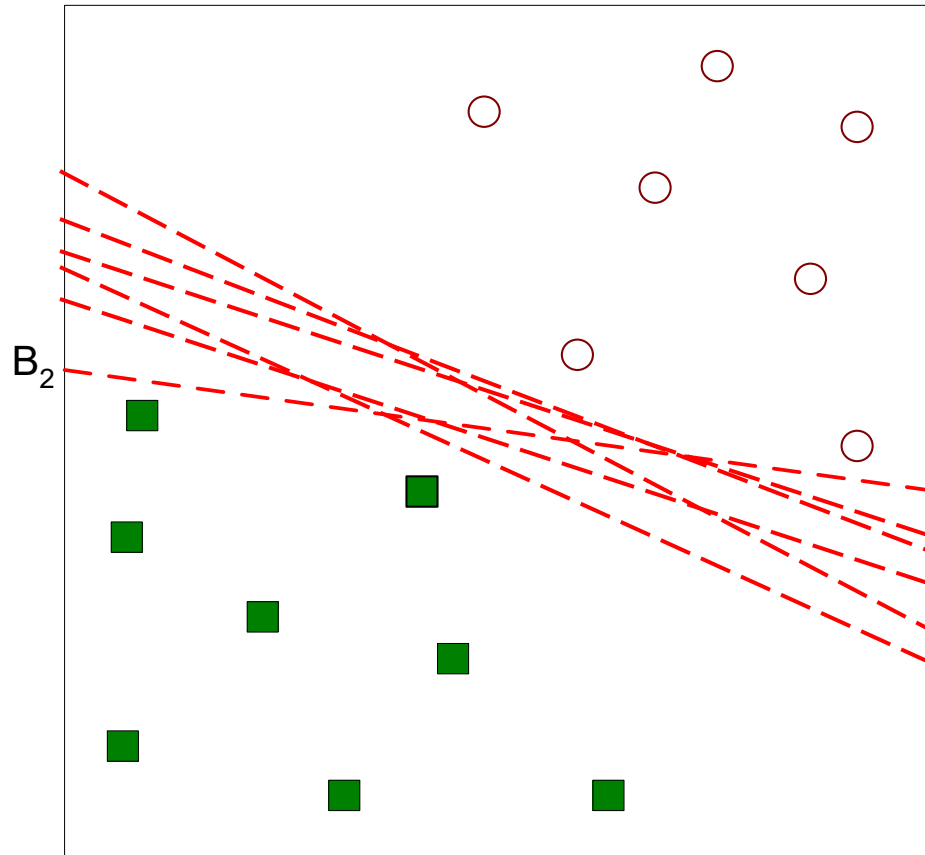
- One Possible Solution

Support Vector Machines (SVM)



- Another possible solution

Support Vector Machines (SVM)



- Other possible solutions

